

Database Systems - MCQ Question Bank

20 Multiple Choice Questions | Answer Key on Last Page

- 1. Which of the following best defines a Database Management System (DBMS)?**
 - A. A collection of application programs only
 - B. Software that enables creation, management, and controlled access to a database
 - C. A type of computer hardware for storing files
 - D. A programming language used to write operating systems
- 2. In the relational model, a table is also referred to as a:**
 - A. Tuple
 - B. Attribute
 - C. Relation
 - D. Domain
- 3. Which key uniquely identifies each record in a table and cannot contain NULL values?**
 - A. Foreign Key
 - B. Candidate Key
 - C. Primary Key
 - D. Alternate Key
- 4. A foreign key in a relational table is used to:**
 - A. Uniquely identify rows within the same table
 - B. Enforce a link between data in two tables
 - C. Speed up query execution
 - D. Store encrypted data
- 5. Which normal form eliminates partial dependency of non-key attributes on the primary key?**
 - A. First Normal Form (1NF)
 - B. Second Normal Form (2NF)
 - C. Third Normal Form (3NF)
 - D. Boyce-Codd Normal Form (BCNF)
- 6. A relation is in First Normal Form (1NF) if:**
 - A. It has no transitive dependency
 - B. All attributes contain only atomic (indivisible) values
 - C. Every determinant is a candidate key
 - D. It has no foreign keys
- 7. Which SQL command is used to remove a table's structure along with all its data?**
 - A. DELETE
 - B. TRUNCATE
 - C. DROP
 - D. REMOVE
- 8. Which SQL clause is used to filter groups after aggregation?**
 - A. WHERE

- B. GROUP BY
- C. HAVING
- D. ORDER BY

9. Which type of SQL JOIN returns all rows from both tables, with NULLs where there is no match?

- A. INNER JOIN
- B. LEFT JOIN
- C. RIGHT JOIN
- D. FULL OUTER JOIN

10. Which of the following is NOT one of the ACID properties of a transaction?

- A. Atomicity
- B. Consistency
- C. Isolation
- D. Availability

11. The property of a transaction that ensures it is treated as a single indivisible unit is called:

- A. Atomicity
- B. Durability
- C. Isolation
- D. Consistency

12. Which of the following best describes a deadlock in a DBMS?

- A. A query that returns no results
- B. Two or more transactions waiting indefinitely for each other to release locks
- C. A failure of the database server hardware
- D. An error caused by invalid SQL syntax

13. Which indexing structure is most commonly used by databases for efficient range queries?

- A. Hash Index
- B. B+ Tree Index
- C. Bitmap Index
- D. Linked List Index

14. Which of the following is a Data Definition Language (DDL) command?

- A. SELECT
- B. UPDATE
- C. CREATE
- D. INSERT

15. In an Entity-Relationship (ER) diagram, a diamond shape typically represents:

- A. An entity
- B. An attribute
- C. A relationship
- D. A primary key

16. Which of the following best describes database normalization?

- A. Increasing data redundancy for faster access
- B. Organizing data to reduce redundancy and improve data integrity
- C. Encrypting sensitive data fields
- D. Converting a database into a flat file

17. A view in SQL is best described as:

- A. A physical copy of a table stored on disk
- B. A virtual table based on the result of a SQL query
- C. A backup of the database
- D. An index used to speed up searches

18. Which concurrency control technique uses locks to prevent conflicting access to data?

- A. Timestamp ordering
- B. Lock-based protocol
- C. Optimistic concurrency control
- D. Multiversion concurrency control

19. Which of the following is an example of a NoSQL database?

- A. MySQL
- B. Oracle
- C. MongoDB
- D. PostgreSQL

20. What is the primary purpose of database normalization's Third Normal Form (3NF)?

- A. To remove multi-valued attributes
- B. To eliminate transitive dependencies between non-key attributes
- C. To ensure every table has a foreign key
- D. To combine multiple tables into one

Answer Key

1. B 2. C 3. C 4. B

5. B 6. B 7. C 8. C

9. D 10. D 11. A 12. B

13. B 14. C 15. C 16. B

17. B 18. B 19. C 20. B